Binary Neutron Stars

Day 1: Relativistic Stellar Structure and the Tolman–Oppenheimer–Volkoff Equation

Comprehensive Lecture Notes with Detailed Discussion

Course arc:

Neutron star formation $\longrightarrow$ Astrophysical observations $\longrightarrow$ Binary mergers $\longrightarrow$ Gravitational-wave signals $\longrightarrow$ Tidal deformability $\longrightarrow$ Equation of state

These notes expand on the Day 1 slides. Boxed material highlights key results, physical interpretation, subtle points, and common misconceptions.

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1 Introduction and Motivation

Neutron stars sit at the intersection of every frontier of modern physics. Their interiors reach densities exceeding nuclear saturation density,

$$\rho_{\text{sat}} \approx 2.8 \times 10^{14}\ \text{g}\,\text{cm}^{-3}, \tag{1}$$

where the microphysics of matter is genuinely unknown. Their gravitational fields are so strong that Newtonian gravity fails completely. And since 2017 we can observe them merging in binaries through gravitational waves, providing a new window onto their interiors.

The goal of this course is to understand the chain:

$$\underbrace{p = p(\epsilon)}_{\text{equation of state}} \xrightarrow{\text{TOV}} \underbrace{M,\, R}_{\text{structure}} \xrightarrow{\text{tidal theory}} \underbrace{\Lambda}_{\text{deformability}} \xrightarrow{\text{GW phase}} \underbrace{\tilde{\Lambda}}_{\text{observable}} .$$

Today we establish the first arrow: given the EOS, how do we compute the mass and radius of a neutron star? That requires the Tolman–Oppenheimer–Volkoff (TOV) equation — the fully relativistic equation of hydrostatic equilibrium.

We distinguish carefully between:

- ϵ — total energy density (used in general relativity),
- ρ — rest-mass density (used in Newtonian limits).

In the non-relativistic limit, $\epsilon \approx \rho c^2$.
Unless otherwise stated, we work in geometrised units $G = c = 1$. Factors of c are restored explicitly when discussing observables.

The three macroscopic observables that link observations to interior physics are:

- **Mass** M — measured precisely from pulsar timing and orbital mechanics.
- **Radius** R — measured (with more uncertainty) from X-ray observations, especially NICER.
- **Tidal deformability** Λ — measured from the gravitational-wave phase in binary mergers.

All three depend on a single fundamental input: the equation of state $p = p(\epsilon)$.

2 Neutron Star Formation

2.1 Core-Collapse Supernovae

Stars with initial masses in the range

$$8\,M_\odot \lesssim M \lesssim 25\,M_\odot \tag{2}$$

end their lives in a **core-collapse supernova**. The evolutionary sequence is:

Massive star → Red supergiant → Core collapse → Supernova → Neutron star.

2.1.1 The Onion-Shell Structure

During the late stages of stellar evolution, successive nuclear burning shells develop from the outside in: hydrogen → helium → carbon → neon → oxygen → silicon → **iron**. This produces a characteristic layered ("onion-shell") structure in the stellar interior.

The iron core is the endpoint of exothermic nuclear burning. Iron-group nuclei sit at the peak of the nuclear binding-energy curve, so *iron fusion does not release energy.* No further burning can provide pressure support against gravity.

2.1.2 The Chandrasekhar Instability

The iron core is supported by **electron degeneracy pressure**. This pressure source has a fundamental limit: the Chandrasekhar mass,

$$M_{\rm Ch} \approx 1.4\, M_\odot. \tag{3}$$

When the core mass reaches $M_{\rm Ch}$, electron degeneracy pressure can no longer support it and *collapse becomes unavoidable.*

The Chandrasekhar limit for a white dwarf composed of iron differs slightly from the classical value for carbon/oxygen white dwarfs because the electron fraction Y_e (electrons per baryon) is lower in iron. During collapse, Y_e decreases further as electrons are captured onto protons. The limit that matters here is the *dynamical* Chandrasekhar mass, which also includes general relativistic corrections.

2.1.3 Collapse Dynamics

Once the core exceeds $M_{\rm Ch}$, collapse proceeds on a free-fall timescale,

$$t_{\rm ff} \sim \frac{1}{\sqrt{G\rho}} \sim 0.1\ \text{s}. \tag{4}$$

This estimate is accurate up to factors of order unity (e.g. $(3\pi/32G\rho)^{1/2}$ for a uniform sphere). During the collapse, three processes occur simultaneously:

1. **Electron capture:** $e^- + p \to n + \nu_e$. Protons absorb electrons and the matter becomes increasingly neutron-rich.
2. **Neutrino trapping:** At densities $\rho \gtrsim 10^{12}\ \text{g}\,\text{cm}^{-3}$, the mean free path of neutrinos becomes shorter than the core radius. Neutrinos are trapped and provide additional pressure during the initial collapse (proto-neutron star phase).
3. **Nuclear repulsion:** At nuclear saturation density, $\rho_{\rm nuc} \sim 2\text{–}3 \times 10^{14}\ \text{g}\,\text{cm}^{-3}$, the strong nuclear force becomes repulsive and halts the collapse. The core "bounces", launching a shock wave outward.

The shock wave stalls in the outer core but is revived — by a combination of neutrino heating, convection, and turbulence — to expel the outer stellar layers as a *supernova explosion.* The inner core becomes the neutron star.

Roughly 3×10^{53} erg of gravitational binding energy are released during the collapse — comparable to the total energy output of the Sun over its entire main-sequence lifetime — and about 99% of this emerges as neutrinos. Only $\sim 1\%$ goes into the kinetic energy of the supernova ejecta. This is why neutrinos were detected from SN 1987A even before the optical brightening reached Earth. The neutrino signal from a future Galactic supernova

would give us real-time data on the proto-neutron star cooling and the equation of state at high temperature.

2.2 Typical Neutron Star Parameters

After the proto-neutron star cools (over $\sim$ 10–100 s by neutrino emission), a cold neutron star is left with the following typical properties:

Property	**Typical value**	**Notes**
Mass	1.2–$2.2\,M_\odot$	Precisely measurable
Radius	10–13 km	Model-dependent, GW/X-ray constrained
Central density	$(5$–$10)\,\rho_{\rm sat}$	Inaccessible directly
Surface gravity	$\sim 10^{14}$ cm s^{-2}	$\sim 10^{11}\times$ Earth
Magnetic field	10^8–10^{15} G	Depends on type
Rotation period	ms to seconds	Recycled vs. young pulsars
Surface temperature	$\sim 10^6$ K	Peaks in soft X-rays

To appreciate the compactness: if a neutron star had the radius of the Earth ($\sim$ 6400 km), it would contain roughly 10^6 Earth masses. A sugar-cube of neutron star matter would weigh $\sim 10^9$ tonnes. The surface gravity is so strong that a 1-metre fall releases energy equivalent to a nuclear weapon.

3 Astrophysical Observations

3.1 Radio Pulsars

The most prolific channel for neutron star detection is through **radio pulsars**: rotating, magnetised neutron stars whose electromagnetic beams sweep across the Earth like a lighthouse.

More than 3 000 radio pulsars are now known. Their timing provides extraordinarily precise measurements of:

- Pulse period P and spin-down rate $\dot{P}$, giving the inferred surface magnetic field $B \sim 3.2 \times 10^{19}\sqrt{P\dot{P}}$ G.
- Neutron star mass via Keplerian orbital parameters and, crucially, post-Keplerian corrections (Shapiro delay, orbital precession, etc.).

3.1.1 The Maximum Mass Constraint

Radio pulsar timing has established a hard lower bound on the maximum neutron star mass:

Pulsar	**Mass** ($M_\odot$)	**Year**
PSR J1614−2230	1.908 ± 0.016	2010
PSR J0348+0432	2.01 ± 0.04	2013
PSR J0952−0607	2.35 ± 0.17	2022

A measurement of $M_{\max} > 2\,M_\odot$ does *not* tell us the maximum possible mass — it places a *lower bound* on it. The actual maximum mass depends on the EOS, which is what we are trying to determine. What the observations do is *rule out* any EOS whose maximum mass falls below the observed value. Many models with hyperons or free quarks predict $M_{\max} \lesssim 1.7\,M_\odot$ and are therefore excluded.

3.2 X-ray Observations

3.2.1 Accreting Neutron Stars

In low-mass X-ray binaries (LMXBs), a neutron star accretes matter from a companion star. The accretion disk reaches temperatures of $\sim 10^7$ K, producing bright X-ray emission. Type I X-ray bursts — thermonuclear runaways on the neutron star surface — can in principle constrain both mass and radius, but systematic uncertainties in the atmosphere modelling are large.

3.2.2 NICER and Pulse Profile Modelling

The **Neutron Star Interior Composition ExploreR** (NICER) instrument on the ISS measures soft X-ray photons from rotating neutron stars with unprecedented timing precision ($\sim$ 100 ns).

The method works as follows:

1. A hot spot on the neutron star surface emits X-rays anisotropically.
2. As the star rotates, the observed flux varies in a way that depends on the star's spacetime geometry.
3. The shape of the pulse profile encodes both M and R simultaneously.

Key NICER results:

- PSR J0030+0451: $M \approx 1.34\,M_\odot$, $R \approx 12.7$ km (Miller et al. 2019).
- PSR J0740+6620: $M \approx 2.08\,M_\odot$, $R \approx 12.4$ km (Riley et al. 2021).

These measurements place direct constraints on the M–R relation independent of gravitational-wave observations.

3.3 Gravitational-Wave Observations

3.3.1 Binary Evolution and Merger

A binary neutron star system loses orbital energy and angular momentum to gravitational radiation. The orbital separation shrinks according to the Peters formula:

$$\frac{da}{dt} = -\frac{64}{5} f(e) \frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 a^3}, \tag{5}$$

leading to merger on a timescale $t_{\rm merge} \propto a_0^4$. The Hulse–Taylor pulsar (PSR B1913+16, discovered 1974) showed orbital decay matching this prediction to 0.2% precision — the first indirect evidence for gravitational waves, awarded the Nobel Prize in 1993.

3.3.2 GW170817

The first direct detection of a binary neutron star merger was **GW170817**, observed on 17 August 2017 by LIGO (Hanford and Livingston) and Virgo.

Property	**Value**
Total mass	$M_{\rm tot} = 2.73^{+0.04}_{-0.01}\, M_\odot$
Chirp mass	$\mathcal{M} = 1.186^{+0.001}_{-0.001}\, M_\odot$
Distance	$d \approx 40$ Mpc
Tidal deformability	$\tilde{\Lambda} \leq 800$ (90% C.L.)
Merger remnant	Unknown (BH or NS)

GW170817 was accompanied by a spectacular set of electromagnetic counterparts, making it the founding event of **multimessenger gravitational-wave astronomy**:

- **GRB 170817A**: A short gamma-ray burst detected by Fermi-GBM and INTEGRAL, ~ 1.7 s after merger. This confirmed the long-suspected association between short GRBs and neutron star mergers.
- **AT2017gfo**: An optical/infrared transient (kilonova) driven by the radioactive decay of r-process elements synthesised in the merger ejecta. The observation confirmed that neutron star mergers are a major site of heavy-element nucleosynthesis (gold, platinum, lanthanides, etc.).
- **X-ray and radio afterglow**: Long-lived emission from the relativistic jet and cocoon, continuing for years.

The tidal constraint from GW170817 translated to a radius bound: $R_{1.4} \lesssim 13.6$ km (at 90% confidence). Combined with the $2\,M_\odot$ pulsar constraint, the allowed EOS band became significantly narrower. However, the precise upper limit on $\tilde{\Lambda}$ depends sensitively on the prior assumed for the mass ratio, the spin, and the EOS parameterisation. Different analyses give somewhat different numbers, so one should quote GW170817 constraints as order-of-magnitude guidance rather than precise boundaries.

3.3.3 GW190425

The second binary neutron star event, **GW190425** (April 2019), had a significantly larger total mass:

$$M_{\rm tot} \sim 3.4\, M_\odot, \tag{6}$$

well above any known Galactic double neutron star system (which have $M_{\rm tot} \lesssim 2.9\, M_\odot$). No electromagnetic counterpart was detected, partly because only one detector (LIGO Livingston) had good sensitivity at the time, giving a poor sky localisation covering ~ 7000 deg^2.

The nature of GW190425 remains debated:

- An unusually massive double neutron star?
- A neutron star – black hole system?
- A product of a different formation channel (e.g. dynamical vs. field binary)?

3.3.4 Future Prospects

Current LVK estimates place the binary neutron star merger rate at

$$\mathcal{R}_{\rm BNS} \sim 10 - 1700\ {\rm Gpc}^{-3}\,{\rm yr}^{-1}, \tag{7}$$

with the central value ~ 300 Gpc^{-3} yr^{-1}. During the current O4 observing run and beyond, tens of detections per year are expected. Third-generation detectors (Einstein Telescope, Cosmic Explorer) will observe *thousands* of mergers per year out to cosmological distances, enabling statistical EOS inference of unprecedented precision.

4 Why Tidal Deformability?

Before deriving the stellar structure equations, it is worth understanding precisely why gravitational waves probe neutron star interiors at all.

During the late inspiral (orbital separation $\lesssim$ few NS radii), the tidal gravitational field of each star deforms its companion. A neutron star responds to an external tidal field $\mathcal{E}_{ij}$ by developing an induced quadrupole moment:

$$Q_{ij} = -\lambda\, \mathcal{E}_{ij}, \tag{8}$$

where λ is the **tidal deformability** (not to be confused with a wavelength). The dimensionless tidal deformability is

$$\Lambda = \frac{\lambda}{M^5} = \frac{2}{3}\, k_2 \left(\frac{R}{M}\right)^5, \tag{9}$$

where k_2 is the dimensionless **tidal Love number** (computed from a perturbation equation — Day 2).

The crucial point is the scaling:

$$\boxed{\Lambda \propto \left(\frac{R}{M}\right)^5.} \tag{10}$$

The fifth power of the compactness ratio means that Λ is an *extraordinarily sensitive* probe of the radius. A 10% change in R (all else equal) changes Λ by $\sim 60\%$. This is why GW observations are so powerful for constraining the EOS.

The induced quadrupole modifies the gravitational potential of the companion, altering the orbital energy. This shows up as an additional phase contribution to the gravitational waveform:

$$\delta\Psi_{\text{tidal}} \propto \tilde{\Lambda} \left(\frac{G M_{\text{tot}} \pi f}{c^3}\right)^{-5/3}, \tag{11}$$

where $\tilde{\Lambda}$ is the mass-weighted combination of the two stars' tidal deformabilities and f is the GW frequency. This phase shift accumulates over the thousands of inspiral cycles visible to LIGO, making it measurable. Stiffer EOS $\rightarrow$ larger R $\rightarrow$ larger Λ $\rightarrow$ earlier phase departure from the point-mass waveform.

5 Assumptions for Stellar Structure

We now develop the relativistic theory of neutron star structure. The starting assumptions are:

1. **Static:** The star is not rotating or oscillating. The metric is time-independent. (Rotation will be treated as a perturbation in a later lecture.)

2. **Spherically symmetric:** The star is spherically symmetric. Combined with the static assumption, this is sometimes called "Schwarzschild symmetry".

3. **Perfect fluid:** The matter inside the star behaves as an ideal fluid, with no viscosity or heat conduction. This is an excellent approximation because:

- viscous forces are negligible compared to pressure gradients in equilibrium,
- thermal gradients in cold neutron stars are tiny compared to pressure gradients.

4. **Cold eos:** We neglect thermal effects. This is valid for mature neutron stars (age $\gg$ cooling timescale $\sim 10^4$ yr) where $kT \ll E_F$ (the Fermi energy). The merger remnant is *hot* and requires a finite-temperature EOS, but for the inspiral phase the cold approximation is excellent.

The assumption of spherical symmetry simplifies the problem enormously. In reality, spinning neutron stars are oblate and require a 2D PDE (instead of 1D ODE) for the structure. However, for all observed pulsars, the oblateness is a small correction:

$$\frac{R_{\rm eq} - R_{\rm pol}}{R} \sim \left(\frac{\Omega}{\Omega_K}\right)^2 \ll 1, \tag{12}$$

where $\Omega_K \sim \sqrt{M/R^3}$ is the Keplerian break-up frequency. Even the fastest-spinning pulsar known (PSR J1748−2446ad, $f = 716$ Hz) has oblateness $\lesssim 10^{-2}$.

6 The Static Spherically Symmetric Metric

6.1 General Form

The most general static, spherically symmetric (SSS) spacetime metric in (t, r, θ, ϕ) coordinates takes the form

$$ds^2 = -e^{\nu(r)}\, dt^2 + e^{\lambda(r)}\, dr^2 + r^2\, d\Omega^2, \tag{13}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ and $\nu(r)$, $\lambda(r)$ are two unknown functions to be determined by the Einstein equations.

Equation (13) is *not* an ansatz — it is a mathematical theorem. For any static, spherically symmetric spacetime, one can always choose coordinates such that the $g_{\theta\theta}$ and $g_{\phi\phi}$ components take the forms r^2 and $r^2 \sin^2\theta$ respectively. This choice defines the coordinate r as the "areal radius": a 2-sphere at coordinate radius r has proper area $4\pi r^2$. The functions ν and λ (or equivalently g_{tt} and g_{rr}) are then determined by the field equations. Off-diagonal terms like g_{tr} vanish by the stationarity and time-reversal symmetry of the static metric.

6.2 The Mass Function

It is convenient to replace $\lambda(r)$ by a new function $m(r)$ defined through

$$e^{-\lambda(r)} = 1 - \frac{2m(r)}{r}. \tag{14}$$

This is a definition, not an approximation. The motivation is that, outside the star, the metric must reduce to the Schwarzschild metric:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2, \tag{15}$$

which corresponds to $e^{-\lambda} = 1 - 2M/r$. So by construction, $m(r)$ interpolates between $m(0) = 0$ (no mass inside a point) and $m(R) = M$ (total gravitational mass).

6.3 Physical Interpretation of ν and λ

- $e^{\nu/2}$ is the **gravitational redshift factor**. A photon emitted with frequency f_e at radius r is observed far away with frequency $f_\infty = f_e\, e^{\nu(r)/2}$.
- $e^{\lambda/2}$ is the **radial stretching factor**. The proper distance between two shells at r and $r + dr$ is $dl = e^{\lambda/2}\, dr$, which is *larger* than the coordinate separation dr inside a massive body. This is a manifestation of spatial curvature.

7 Stress-Energy Tensor and Einstein Equations

7.1 Perfect Fluid Stress-Energy Tensor

For a perfect fluid, the stress-energy tensor is

$$T^{\mu\nu} = (\epsilon + p)\, u^\mu u^\nu + p\, g^{\mu\nu}, \tag{16}$$

where:

- ϵ = total energy density (rest mass + internal energy + kinetic energy of constituents),
- p = isotropic pressure,
- u^μ = four-velocity of the fluid element.

ϵ is not the rest-mass density ρ. At supranuclear densities these differ significantly. The quantity that appears in the EOS is the *total* energy density ϵ, which in terms of the rest-mass density ρ and specific internal energy e is:

$$\epsilon = \rho + \rho e, \tag{17}$$

with ($G = c = 1$). In the nonrelativistic limit $\rho e \ll \rho$, so $\epsilon \approx \rho$ and the two are interchangeable. For neutron stars this approximation fails badly in the deep interior.

7.2 Four-Velocity for Static Matter

For a static fluid element (as required by our assumptions), $u^r = u^\theta = u^\phi = 0$. The normalisation condition $g_{\mu\nu}u^\mu u^\nu = -1$ gives:

$$g_{tt}(u^t)^2 = -1 \quad \Rightarrow \quad -e^\nu (u^t)^2 = -1 \quad \Rightarrow \quad u^t = e^{-\nu/2}, \tag{18}$$

so $u^\mu = (e^{-\nu/2}, 0, 0, 0)$.

7.3 The Einstein Field Equations

We work in geometrised units $G = c = 1$. The Einstein field equations are

$$G_{\mu\nu} = 8\pi T_{\mu\nu}. \tag{19}$$

For the SSS metric and the perfect fluid stress-energy tensor, the independent non-trivial components are the tt and rr equations, plus the contracted Bianchi identity $\nabla_\mu T^{\mu\nu} = 0$.

8 The Mass Equation

8.1 Derivation

Computing the Einstein tensor $G_{\mu\nu}$ for the metric (13) and evaluating the tt component of $G_{\mu\nu} = 8\pi T_{\mu\nu}$, one finds (after some algebra):

$$\frac{d}{dr}\left(re^{-\lambda}\right) = 1 - 8\pi r^2 \epsilon. \tag{20}$$

Using the definition (14), this becomes:

$$\boxed{\frac{dm}{dr} = 4\pi r^2 \epsilon.} \tag{21}$$

This is the **mass equation**.

8.2 Physical Interpretation

Equation (21) looks like the Newtonian formula for the mass enclosed in a sphere:

$$dm = \underbrace{4\pi r^2\, dr}_{\text{shell volume}} \times \underbrace{\epsilon}_{\text{energy density}}.$$

This analogy is *misleading*. The "proper volume" of the shell at radius r is $dV_{\text{proper}} = 4\pi r^2 e^{\lambda/2}\, dr$, which differs from $4\pi r^2\, dr$ by the radial stretching factor $e^{\lambda/2} > 1$. The gravitational mass $m(r)$ is *not* simply the integral of the energy density over the proper volume. The difference between $m(R)$ and the integral of ϵ over the proper volume is the (negative) gravitational binding energy. In GR, Gravitational binding energy reduces the total mass seen from infinity; a bound system weighs less than its separated parts.

9 Derivation of the TOV Equation

9.1 The Hydrostatic Equilibrium Condition

Energy-momentum conservation requires

$$\nabla_\mu T^{\mu\nu} = 0. \tag{22}$$

The $\nu = r$ component gives the radial force balance:

$$\frac{dp}{dr} = -\frac{1}{2}(\epsilon + p)\, \nu'(r), \tag{23}$$

where $\nu'(r) = d\nu/dr$. This equation has a clear structure: the pressure gradient must balance a "gravitational force" proportional to $(\epsilon + p)\nu'$. The factor of $(\epsilon + p)$ already signals a departure from Newtonian gravity, where only the rest-mass density appears.

9.2 Computing ν' from the Einstein Equations

The rr component of $G_{\mu\nu} = 8\pi T_{\mu\nu}$ gives:

$$e^{-\lambda}\left(\frac{\nu'}{r} + \frac{1}{r^2}\right) - \frac{1}{r^2} = 8\pi p. \tag{24}$$

Substituting $e^{-\lambda} = 1 - 2m/r$ and solving for ν':

$$\nu' = \frac{2(m + 4\pi r^3 p)}{r(r - 2m)}. \tag{25}$$

Derivation of equation (25)

Starting from the rr Einstein equation:

$$e^{-\lambda}\left(\frac{\nu'}{r} + \frac{1}{r^2}\right) - \frac{1}{r^2} = 8\pi p \tag{26}$$

$$\left(1 - \frac{2m}{r}\right)\frac{\nu'}{r} + \frac{1}{r^2}\left(1 - \frac{2m}{r} - 1\right) = 8\pi p \tag{27}$$

$$\left(1 - \frac{2m}{r}\right)\frac{\nu'}{r} - \frac{2m}{r^3} = 8\pi p \tag{28}$$

$$\nu' = \frac{r}{1 - 2m/r}\left(\frac{2m}{r^3} + 8\pi p\right) \tag{29}$$

$$= \frac{2m/r^2 + 8\pi r p}{1 - 2m/r} \tag{30}$$

$$= \frac{2(m + 4\pi r^3 p)}{r(r - 2m)}. \tag{31}$$

9.3 The TOV Equation

Substituting (25) into (23):

$$\frac{dp}{dr} = -\frac{(\epsilon + p)(m + 4\pi r^3 p)}{r(r - 2m)}. \tag{32}$$

Together with the mass equation,

$$\frac{dm}{dr} = 4\pi r^2 \epsilon, \tag{33}$$

and an equation of state $p = p(\epsilon)$, this constitutes a closed system of ODEs determining the stellar structure.

10 Physical Interpretation of the TOV Equation

It is instructive to compare the TOV equation with its Newtonian counterpart to identify the three general-relativistic corrections.

10.1 The Newtonian Limit

In the Newtonian limit, $p \ll \epsilon$ and $2m/r \ll 1$. In the Newtonian limit, $\epsilon \approx \rho c^2$ also. Applying these two conditions to (32):

$$\frac{dp}{dr} \approx -\frac{\rho \cdot m}{r \cdot r} = -\frac{\rho m}{r^2}. \tag{34}$$

This is precisely the **Newtonian hydrostatic equilibrium equation**. The TOV equation is thus the fully relativistic generalisation.

10.2 The Three Relativistic Correction Terms

Rewrite the TOV equation schematically:

$$\frac{dp}{dr} = -\frac{\epsilon m}{r^2} \underbrace{\left(1 + \frac{p}{\epsilon}\right)}_{(1)} \underbrace{\left(1 + \frac{4\pi r^3 p}{m}\right)}_{(2)} \underbrace{\left(1 - \frac{2m}{r}\right)^{-1}}_{(3)}. \tag{35}$$

Each factor represents a distinct physical effect absent in Newtonian gravity:

1. **Pressure gravitates:** The factor $(1 + p/\epsilon)$ in the numerator comes from $(\epsilon + p)$ in the stress-energy tensor. In GR, pressure is not "free" — it contributes to $T^{\mu\nu}$ and therefore acts as a source of gravity. Trying to support the star with higher pressure paradoxically makes the gravitational pull stronger. This is the central mechanism behind the maximum mass.

2. **Pressure in the mass term:** The factor $(1 + 4\pi r^3 p/m)$ appears because the effective gravitational mass inside radius r includes not just the energy density ϵ but also the pressure p (through the T^{rr} component of the stress-energy tensor). This is the **active gravitational mass** effect.

3. **Spacetime curvature:** The factor $(1 - 2m/r)^{-1}$ in the denominator is the Schwarzschild redshift factor. As the star becomes more compact ($2m/r \to 1$), this factor diverges, enormously enhancing the effective gravitational pull. This is the typical case for a black hole but in neutron star the limit is never reached. Near the Buchdahl limit (Section 14), this term becomes the dominant relativistic correction.

All three factors are ≥ 1 and all three *increase* the magnitude of dp/dr relative to the Newtonian case. GR makes gravity *stronger* inside a compact star, which is why there is a maximum mass in GR but not in Newtonian gravity with a non-degenerate gas.

A very common misconception: "Pressure supports the star against gravity, so more pressure means more stability." In GR this is simply wrong. Increasing the central pressure increases both the pressure gradient (which supports the star) *and* the effective gravitational pull (through correction terms 1 and 2 above). Beyond the maximum mass, the gravitational increase wins and no hydrostatic equilibrium is possible.

11 Integration Procedure and Boundary Conditions

11.1 Boundary Conditions

The system of ODEs (21)–(32) requires boundary conditions. At the centre:

$$m(0) = 0, \qquad p(0) = p_c \ \text{(central pressure)}. \tag{36}$$

The central pressure (or equivalently, the central density ϵ_c) is the single free parameter of the problem.

At the surface, the boundary condition is:

$$p(R) = 0, \tag{37}$$

where R is the stellar radius. The total gravitational mass is $M = m(R)$.

Near $r = 0$, regularity requires:

$$m(r) \approx \tfrac{4}{3}\pi r^3 \epsilon_c + O(r^5), \tag{38}$$

so $m/r^2 \approx (4\pi/3)\, r\, \epsilon_c \to 0$ as $r \to 0$. The TOV equation is therefore **regular** at $r = 0$: there is no coordinate singularity. In numerical integration, one typically seeds the solution with a Taylor expansion near the centre:

$$m(r) \approx \tfrac{4}{3}\pi r^3 \epsilon_c, \tag{39}$$

$$p(r) \approx p_c - \tfrac{2}{3}\pi(\epsilon_c + p_c)(\epsilon_c + 3p_c)\, r^2, \tag{40}$$

and then integrates outward using a standard ODE solver (e.g. Runge–Kutta 4th order) until $p \to 0$.

11.2 The Integration Scheme

The full scheme for computing one stellar model is:

1. Choose a central energy density ϵ_c.
2. Compute the central pressure $p_c = p(\epsilon_c)$ from the EOS.
3. Initialise: $m_0 = 0$, $p_0 = p_c$, at small $r = r_0 \sim 1$ m.
4. Integrate the coupled system

$$\frac{dm}{dr} = 4\pi r^2 \epsilon, \qquad \frac{dp}{dr} = -\frac{(\epsilon + p)(m + 4\pi r^3 p)}{r(r - 2m)},$$

   using $\epsilon = \epsilon(p)$ from the inverted EOS.
5. Stop when $p(R) = 0$ (or below a numerical threshold corresponding to the crust surface density $\sim 10^{11}\ \mathrm{g\,cm^{-3}}$).
6. Record $M = m(R)$ and R.

By repeating this for many values of ϵ_c, one traces out the M–R curve.

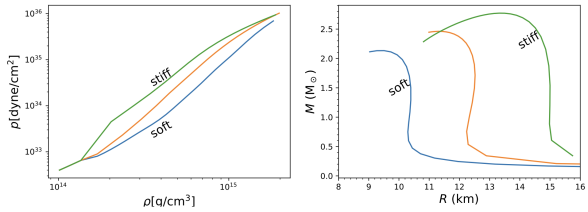

Figure 1: Schematic mass–radius (M–R) relation for neutron stars. Each curve corresponds to a different equation of state. The maximum mass marks the onset of instability ($dM/d\epsilon_c < 0$)

12 The Mass–Radius Relation

12.1 Structure of the M–R Curve

The M–R relation has the following general features:

- At low central densities($\epsilon_c \sim \rho_{\text{sat}}$), the mass increases with ϵ_c, while the radius typically decreases or varies weakly depending on the EOS.
- The curve reaches a **maximum mass** M_{max} at some critical central density.
- Beyond M_{max}, adding more mass is impossible: the star would collapse to a black hole.

12.2 The Stability Criterion

A neutron star is dynamically stable if and only if

$$\frac{dM}{d\epsilon_c} > 0. \tag{41}$$

Stars on the branch where $dM/d\epsilon_c < 0$ are unstable to radial perturbations and will either collapse to a black hole or explode.

The stability criterion $dM/d\epsilon_c > 0$ is sometimes stated incorrectly as "higher central density always means more massive star". This is true only on the *stable branch*. On the unstable branch (beyond M_{max}), increasing ϵ_c actually *decreases* M — these configurations are not observed in nature.

12.3 Dependence on the Equation of State

The M–R curve is a **one-to-one map** from the EOS: different equations of state produce qualitatively similar curve shapes but with systematically different values of M_{max}, R, and curvature. This means that measuring a few points on the M–R plane directly constrains the EOS.

13 Stiff vs. Soft Equations of State

13.1 Definitions

The stiffness of an EOS is characterised by how rapidly pressure increases with density. Formally, the **speed of sound** in the matter is

$$c_s^2 = \frac{dp}{d\epsilon}. \tag{42}$$

This is a local property of the equation of state. Additional global constraints (e.g. stability and causality of the full stellar configuration) must also be satisfied.

- A **stiff eos** has large c_s^2: pressure increases rapidly with density.
- A **soft eos** has small c_s^2: pressure increases slowly.

A fundamental constraint from causality is $c_s \leq c$, i.e. $dp/d\epsilon \leq 1$ in geometrised units.

13.2 Polytropic Models

A useful toy model is the **polytropic eos**:

$$p = K\rho^\Gamma, \tag{43}$$

where K is a constant and Γ is the adiabatic index. Stiffness is controlled by Γ:

Γ	**Physical regime**	**Stiffness**
4/3	Ultra-relativistic degenerate gas	Soft
5/3	Non-relativistic degenerate gas (white dwarf)	Medium
2–3	Neutron star interior models	Stiff

The radius of a polytropic star scales roughly as

$$R \sim K^{1/(3\Gamma-4)}\, M^{(\Gamma-2)/(3\Gamma-4)}, \tag{44}$$

showing that larger Γ (stiffer EOS) generally produces larger radii.

13.3 Consequences for Neutron Star Observables

Observable	**Stiff eos**	**Soft eos**
Radius R	Larger	Smaller
Maximum mass $M_{\max}$	Higher	Lower
Tidal deformability Λ	Larger	Smaller
Compactness $C = M/R$	Smaller	Larger

Why does stiffness increase $M_{\max}$?
To support a more massive star, you need stronger pressure support at higher densities. A stiff EOS provides this by having a large pressure even at high densities. A soft EOS "gives in" — the pressure does not grow fast enough to resist the relativistic enhancement of gravity.

Why does stiffness increase R?
Stiff matter resists compression. Given the same mass, a star made of stiff matter sits in a wider, puffier configuration. Soft matter compresses more easily, leading to a smaller, denser star.

Observational connection:

The 2017 measurement of GW170817 constrained $\tilde{\Lambda} \lesssim 800$, ruling out the stiffest EOS models (which predicted $\tilde{\Lambda} \sim 3000$–$5000$). The 2010–2022 measurements of $M > 2\,M_\odot$ pulsars ruled out the softest models. The allowed EOS band is now significantly narrower than a decade ago.

13.4 Physical Origins of Stiffness

The microscopic origin of EOS stiffness is an active area of research:

- **Sources of stiffness:** Strong repulsive nuclear interactions at short range, three-body repulsive forces, stiff quark matter (if present).
- **Sources of softness:** Hyperon formation (hyperons add new degrees of freedom, diluting the neutron Fermi pressure — the "hyperon puzzle"), meson condensates (pion or kaon), phase transitions to quark matter.

The lightest hyperons (Λ, Σ^-) become energetically favourable at densities $\gtrsim 2\,\rho_{\rm sat}$. Adding hyperons softens the EOS because they provide new channels for matter to reduce its energy. However, many simpler hyperon models predict $M_{\rm max} \lesssim 1.7\,M_\odot$, which is excluded by the observed $2\,M_\odot$ pulsars. However, $M_{\rm max} \sim 2\,M_\odot$ can be reached with other models. This "hyperon puzzle" remains unsolved: either hyperons are somehow suppressed by repulsive hyperon–nucleon interactions we do not yet understand, or the EOS stiffens again at higher density (e.g. via quark matter), or there is some other physics we are missing.

14 Compactness and the Buchdahl Bound

14.1 Compactness

The **compactness** of a neutron star is defined as

$$C = \frac{M}{R} \qquad (G = c = 1). \tag{45}$$

In physical units, $C = GM/(Rc^2)$. Typical neutron stars have $C \approx 0.15$–0.30.

For comparison:

Object	$C = GM/Rc^2$	**Notes**
Sun	2.1×10^{-6}	Newtonian gravity excellent
White dwarf	$\sim 10^{-4}$	Newtonian, mildly post-Newtonian
Typical NS	0.15–0.30	Strongly relativistic
Buchdahl limit	$4/9 \approx 0.444$	Theoretical maximum for fluid star
Black hole	$1/2$	Horizon at $r_s = 2M$

14.2 The Buchdahl Bound

One of the most elegant results in relativistic stellar theory is the **Buchdahl bound** (1959):

For any static, spherically symmetric, isotropic fluid star with positive energy density and density non-increasing outward ($d\rho/dr \leq 0$), the compactness must satisfy:

$$\frac{2M}{R} < \frac{8}{9}, \qquad \text{equivalently} \quad C < \frac{4}{9} \approx 0.444. \tag{46}$$

14.3 Derivation Sketch

Buchdahl bound (constant density)

Assume $\epsilon = \epsilon_0$. Then

$$m(r) = \frac{4\pi}{3}\epsilon_0 r^3, \qquad M = \frac{4\pi}{3}\epsilon_0 R^3. \tag{47}$$

The TOV equation gives

$$\frac{dp}{dr} = -\frac{(\epsilon_0 + p)(m + 4\pi r^3 p)}{r(r - 2m)}. \tag{48}$$

Separating variables and integrating to the surface ($p(R) = 0$):

$$\frac{1}{3}\frac{p_c + \epsilon_0}{p_c + \epsilon_0/3} = \sqrt{1 - \frac{2M}{R}}. \tag{49}$$

Thus

$$p_c = \epsilon_0 \frac{1 - \sqrt{1 - 2M/R}}{3\sqrt{1 - 2M/R} - 1}. \tag{50}$$

Divergence of p_c implies

$$3\sqrt{1 - \frac{2M}{R}} = 1 \;\Rightarrow\; \boxed{\frac{2M}{R} < \frac{8}{9}}. \tag{51}$$

The constant-density model used above saturates the Buchdahl bound. The full theorem applies more generally to any configuration with monotonically decreasing density and isotropic pressure, but requires a more general argument beyond this special case.

The general proof follows similar reasoning but does not assume constant density. The key result is that the central pressure diverges at $2M/R = 8/9$, establishing the Buchdahl bound. No physical star can have infinite central pressure; therefore $2M/R < 8/9$ for any stable configuration.

The Buchdahl bound can be understood intuitively: as the compactness approaches $8/9$, the denominator $(r - 2m)$ in the TOV equation becomes very small throughout the star. The gravitational term in the TOV equation is enhanced by $(1 - 2m/r)^{-1} \to \infty$, requiring an enormous pressure gradient to maintain equilibrium. The central pressure must diverge at the bound. Since no physical EOS can provide infinite pressure, the star collapses before reaching this limit.

The bound rests on three key assumptions:

1. The energy density is positive ($\epsilon \geq 0$).
2. The density is non-increasing outward ($d\rho/dr \leq 0$).
3. The fluid is isotropic ($p_r = p_t$).

Violating any of these can in principle allow more compact objects. For example, anisotropic pressure models (where $p_r \neq p_t$, sometimes motivated by phase transitions or magnetic fields) can circumvent the Buchdahl bound in theory. Whether such configurations are physically realisable is an open question. Real neutron stars satisfy all three conditions to excellent approximation.

14.4 Newtonian Limit

In the weak-gravity regime,

$$\frac{2M}{R} \ll 1, \tag{52}$$

and pressures are small compared to energy density ($p \ll \epsilon$). The TOV equation reduces to the Newtonian hydrostatic equilibrium equation:

$$\frac{dp}{dr} \approx -\frac{\rho\, m(r)}{r^2}. \tag{53}$$

For a constant-density star, the central pressure is finite and scales as

$$p_c \sim \epsilon_0 \frac{M}{R}. \tag{54}$$

In Newtonian gravity, pressure acts only as a supporting force. Increasing the central pressure always helps stabilise the star. There is no intrinsic upper bound on compactness: one can, in principle, support arbitrarily compact configurations by increasing the pressure.

14.5 General Relativistic Behaviour

In full general relativity, the central pressure is

$$p_c = \epsilon_0 \frac{1 - \sqrt{1 - 2M/R}}{3\sqrt{1 - 2M/R} - 1}. \tag{55}$$

As the compactness increases:

- the numerator increases,
- the denominator decreases,
- the required central pressure grows rapidly.

At the critical compactness

$$\frac{2M}{R} \to \frac{8}{9}, \tag{56}$$

the denominator vanishes and

$$p_c \to \infty. \tag{57}$$

In general relativity, pressure does not only support the star — it also contributes to gravity. Increasing the pressure strengthens the gravitational field, leading to a feedback loop:

$$\text{more pressure} \rightarrow \text{stronger gravity} \rightarrow \text{need more pressure.}$$

This feedback causes the central pressure to diverge at a finite compactness, producing the Buchdahl bound.

14.6 Key Difference

	Newtonian	**General Relativity**
Role of pressure	Supports star	Supports *and* gravitates
Central pressure	Finite	Diverges at $2M/R = 8/9$
Maximum compactness	None	Buchdahl bound
Stability	Always possible (idealised)	Breaks at high compactness

The emergence of a maximum compactness is a purely relativistic effect. It arises from the nonlinear coupling between pressure and gravity in the Einstein equations and has no analogue in Newtonian gravity.

14.7 The Gap Between Buchdahl Limit and Black Holes

There is a finite gap between the most compact possible fluid star and the Schwarzschild radius:

$$\underbrace{\frac{2M}{R} \leq \frac{8}{9}}_{\text{Buchdahl}} < \underbrace{\frac{2M}{R} = 1}_{\text{Schwarzschild horizon}} . \tag{58}$$

No stable fluid star can exist in the range $8/9 < 2M/R \leq 1$. Objects in this range would have to be black holes (or some exotic horizonless compact object, like gravastars, which are theoretically speculative and not observationally established).

This gap is important because it means that if we observe an object with compactness $C > 4/9$, it must be a black hole (or something even more exotic than a fluid star).

15 Surface Gravitational Redshift

Photons emitted from the neutron star surface experience a **gravitational redshift**. A photon emitted at the surface with frequency f_e is observed far away with frequency f_∞:

$$\frac{f_\infty}{f_e} = \sqrt{g_{tt}(R)} = e^{\nu(R)/2} = \left(1 - \frac{2M}{R}\right)^{1/2} . \tag{59}$$

The surface redshift is defined as

$$z = \frac{f_e - f_\infty}{f_\infty} = \left(1 - \frac{2M}{R}\right)^{-1/2} - 1. \tag{60}$$

Example 15.1. For a canonical neutron star with $M = 1.4\,M_\odot$ and $R = 12$ km, the compactness is $C \approx 0.17$, and:

$$z = (1 - 2 \times 0.17)^{-1/2} - 1 = (0.66)^{-1/2} - 1 \approx 1.23 - 1 = 0.23.$$

Spectral lines from the neutron star surface appear shifted by $\approx 23\%$ to longer wavelengths (lower energies). This is a strong-gravity effect that would be completely undetectable from the surfaces of white dwarfs ($z \sim 10^{-4}$) or the Sun ($z \sim 2 \times 10^{-6}$).

In principle, observing spectral lines from the NS surface and identifying them with known atomic transitions would give a direct measurement of $z = (1 - 2C)^{-1/2} - 1$, and therefore of the compactness $C = M/R$. In practice this is extremely difficult:

- The NS atmosphere is so hot that only the lightest elements (H, He) survive — and their spectral lines are in the UV/soft X-ray, where interstellar absorption is severe.
- There is a degeneracy between z and the Doppler shift from the NS rotation.
- The atmospheric modelling introduces significant systematic uncertainties.

Some isolated neutron stars (e.g. the "Magnificent Seven" thermally emitting X-ray sources) may offer cleaner opportunities for such measurements in future observations.

16 Connection to Tidal Deformability and Day 2

Today we have solved the **equilibrium problem**: given an EOS, we can compute M and R by integrating the TOV equation.

The **tidal problem** (Day 2) asks: given an equilibrium star, how does it respond to an external tidal perturbation? The answer requires solving a second set of ODEs (the tidal perturbation equations) atop the background TOV solution:

$$\frac{d\eta}{dr} = F(r, m, p, \epsilon, \eta), \tag{61}$$

where $\eta(r)$ is a function encoding the radial profile of the tidal deformation. At the surface, $\eta(R)$ determines the tidal Love number k_2, and hence the tidal deformability $\Lambda = (2/3)k_2(R/M)^5$.

The tidal Love number k_2 depends on both the compactness C and the interior structure of the star (through $\eta(R)$):

$$k_2 = \frac{8C^5}{5}(1 - 2C)^2\Big[2 + 2C(\eta_R - 1) - \eta_R\Big]\{\ldots\}^{-1}, \tag{62}$$

where $\eta_R = r\,d\ln H/dr\big|_{r=R}$ and $H(r)$ is the even-parity perturbation function. The full expression for k_2 will be derived in Lecture 2. Note that $k_2 \to 0$ as $C \to 1/2$ (for a black hole).

17 Summary and Outlook

17.1 What We Have Established

In this lecture we have:

1. **Motivated the problem** through the chain from neutron star formation to gravitational-wave observations of binary mergers.
2. **Constructed the SSS metric** and defined the mass function $m(r)$ as the natural relativistic generalisation of the enclosed Newtonian mass.
3. **Derived the TOV equation** from the Einstein equations plus energy-momentum conservation, identifying three purely relativistic corrections: pressure gravitates, pressure enters the effective mass, and curvature enhances gravity.
4. **Showed the Newtonian limit**: TOV $\to$ classical hydrostatic equilibrium when $p \ll \epsilon$ and $2m/r \ll 1$.
5. **Discussed the mass–radius relation**: each EOS maps to a unique M–R curve; the stability condition $dM/d\epsilon_c > 0$ separates stable from unstable branches.
6. **Introduced stiff vs. soft eos**: stiff $\to$ large R, large $M_{\max}$, large Λ; soft $\to$ the opposite.
7. **Derived the Buchdahl bound** $2M/R < 8/9$: the fundamental upper limit on compactness for a physical fluid star.
8. **Computed the surface redshift** and related it to compactness.

17.2 Key Equations

Name	Equation	Section
Metric	$ds^2 = -e^{\nu}dt^2 + e^{\lambda}dr^2 + r^2 d\Omega^2$	6
Mass function	$e^{-\lambda} = 1 - 2m(r)/r$	6
Mass equation	$dm/dr = 4\pi r^2 \epsilon$	8
TOV equation	$dp/dr = -(\epsilon + p)(m + 4\pi r^3 p)/[r(r - 2m)]$	9
Newtonian limit	$dp/dr = -\rho m/r^2$	10
Buchdahl bound	$2M/R < 8/9$	14
Surface redshift	$z = (1 - 2M/R)^{-1/2} - 1$	15
Tidal deformability	$\Lambda = (2/3)k_2(R/M)^5$	16

17.3 Conceptual Questions for Self-Study

Exercise 17.1. Why is the Newtonian maximum mass (Chandrasekhar mass) not relevant for neutron stars? What is the *relativistic* mechanism that produces a maximum mass, and how does it differ from the Chandrasekhar instability?

Exercise 17.2. Consider two equations of state: EOS A with $p = K\rho^{5/3}$ (non-relativistic degeneracy) and EOS B with $p = K\rho^{4/3}$ (ultra-relativistic degeneracy). Which is stiffer? What does this predict for the relative sizes of their neutron stars?

Exercise 17.3. The Buchdahl bound assumes $d\rho/dr \leq 0$. Construct a thought experiment where violating this condition might allow a more compact stable configuration. Is such a configuration physically reasonable?

Exercise 17.4. Verify the surface redshift for $M = 1.4\,M_\odot$, $R = 12$ km. What would the redshift be for the most compact realistic neutron star ($C \approx 0.30$)? How does this compare with the Buchdahl limit ($C = 4/9$)?

Exercise 17.5. GW170817 measured $\tilde{\Lambda} \lesssim 800$. Given that $\Lambda \propto (R/M)^5$, estimate the rough upper bound this places on the radius of a $1.4\,M_\odot$ neutron star. (Use $k_2 \approx 0.07$ as a typical value.)